PROCESS_LINEAR

1

Collect data

Gather raw samples from sensors and logs

2

Clean

Remove noise

3

Train

Fit the model to the cleaned data using gradient descent

4

Deploy

Ship it

Key takeaway: **Key point here.**

PROCESS_CYCLIC

1

Plan

Define goals

2

Do

Execute

3

Check

Measure

4

Act

Adjust

5

Repeat

Loop again

↻ and the cycle repeats

WORKED_EXAMPLE

1

Write the equation

$y = mx + b$

2

Substitute values

$m = 2, b = 3, x = 4$

3

Compute

$y = 2*4+3 = 11$

TIMELINE

1950
Turing test proposed

1956
Dartmouth workshop coins AI

1997
Deep Blue beats Kasparov in chess

2012
AlexNet wins ImageNet

2017
Transformers introduced

Key takeaway: **Key point here.**

COMPARE_TABLE

Feature	SQL	NoSQL
Schema	Fixed	Flexible
Scaling	Vertical	Horizontal and elastic across many nodes
Joins	Native	Manual
ACID	Yes	Varies

Key takeaway: **Key point here.**

COMPARE_VISUAL

ADVANTAGES

- Fast to train
- Interpretable
- Low memory

TRADE-OFFS

- Overfits easily on small data and noisy features
- Needs tuning

MATRIX_GRID

High impact / low effort

- Quick wins
- Automate

High / high

- Major projects

Low / low

- Fill-ins

Low / high

- Avoid

REFERENCES

1. Goodfellow et al., Deep Learning, MIT Press 2016
2. Bishop, Pattern Recognition and Machine Learning, Springer 2006
3. Russell and Norvig, AI: A Modern Approach
4. Murphy, Machine Learning: A Probabilistic Perspective
5. Hastie et al., Elements of Statistical Learning

LAYERS

Application What users see

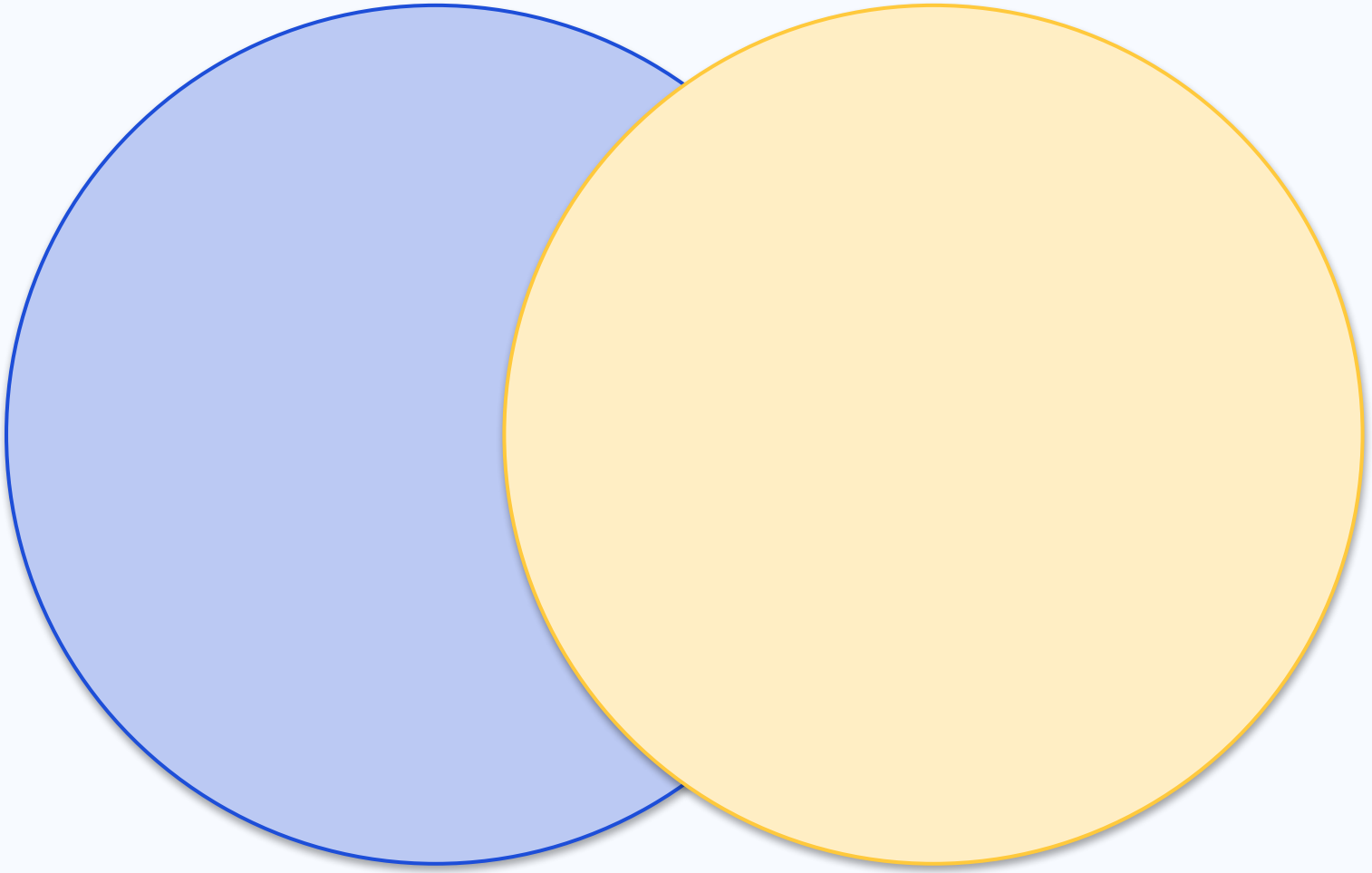
Transport End to end delivery

Network Routing

Link Frames on wire

A short note about the slide topic.

SET_DIAGRAM



- Supervised learning
- Unsupervised learning
- Both use data
- Both need evaluation

STAT_TILES

92%

ACCURACY

On held-out test set

3.1M

USERS

Monthly active

14ms

LATENCY

p95

Key takeaway: **Key point here.**

CAUSE_EFFECT

1

Rising temperature

CO_2 traps heat

2

Ice melts

Sea levels rise

3

Less reflection

Darker surface absorbs more

↺ and the effect feeds back into the cause

FORMULA_CALLOUT

BAYES THEOREM

The probability of a hypothesis given evidence

$$P(A|B) = P(B|A)\,P(A) \,/\, P(B)$$

- P(A) is the prior
- P(B|A) is the likelihood of the evidence given the hypothesis

Overfitting

When a model memorises noise in the training set and fails to generalise.

High variance

Low training error

High test error

Key takeaway: **Key point here.**

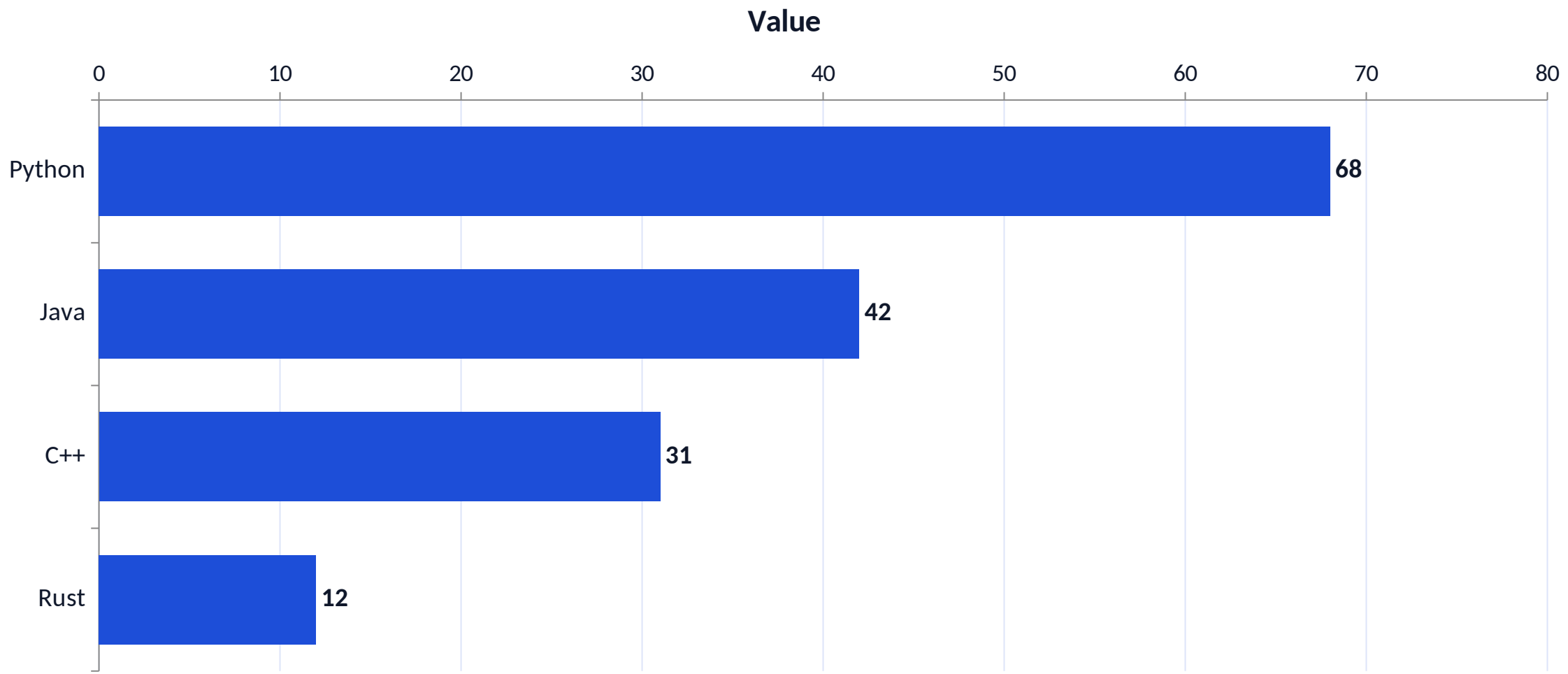
QUOTE

> “The best way to predict the future is to invent it.”
>
> — Alan Kay

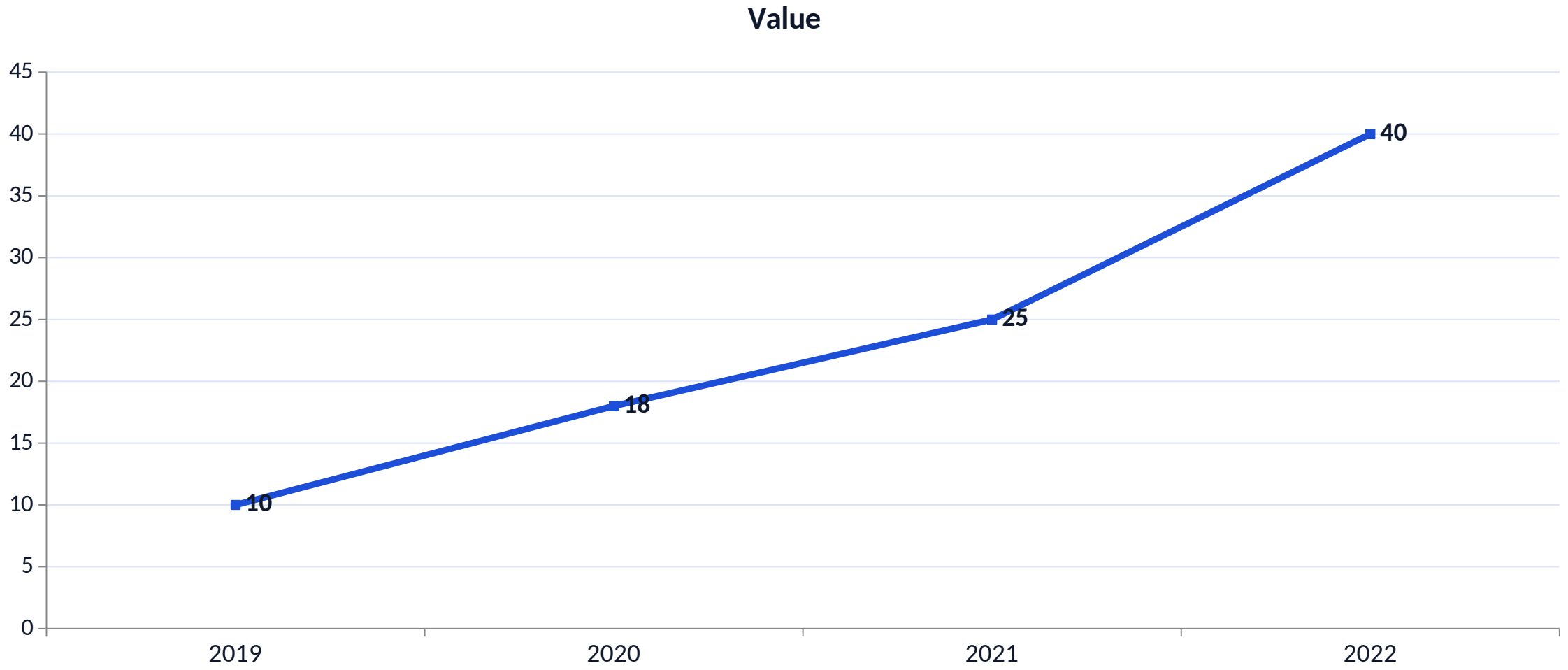
FUNNEL



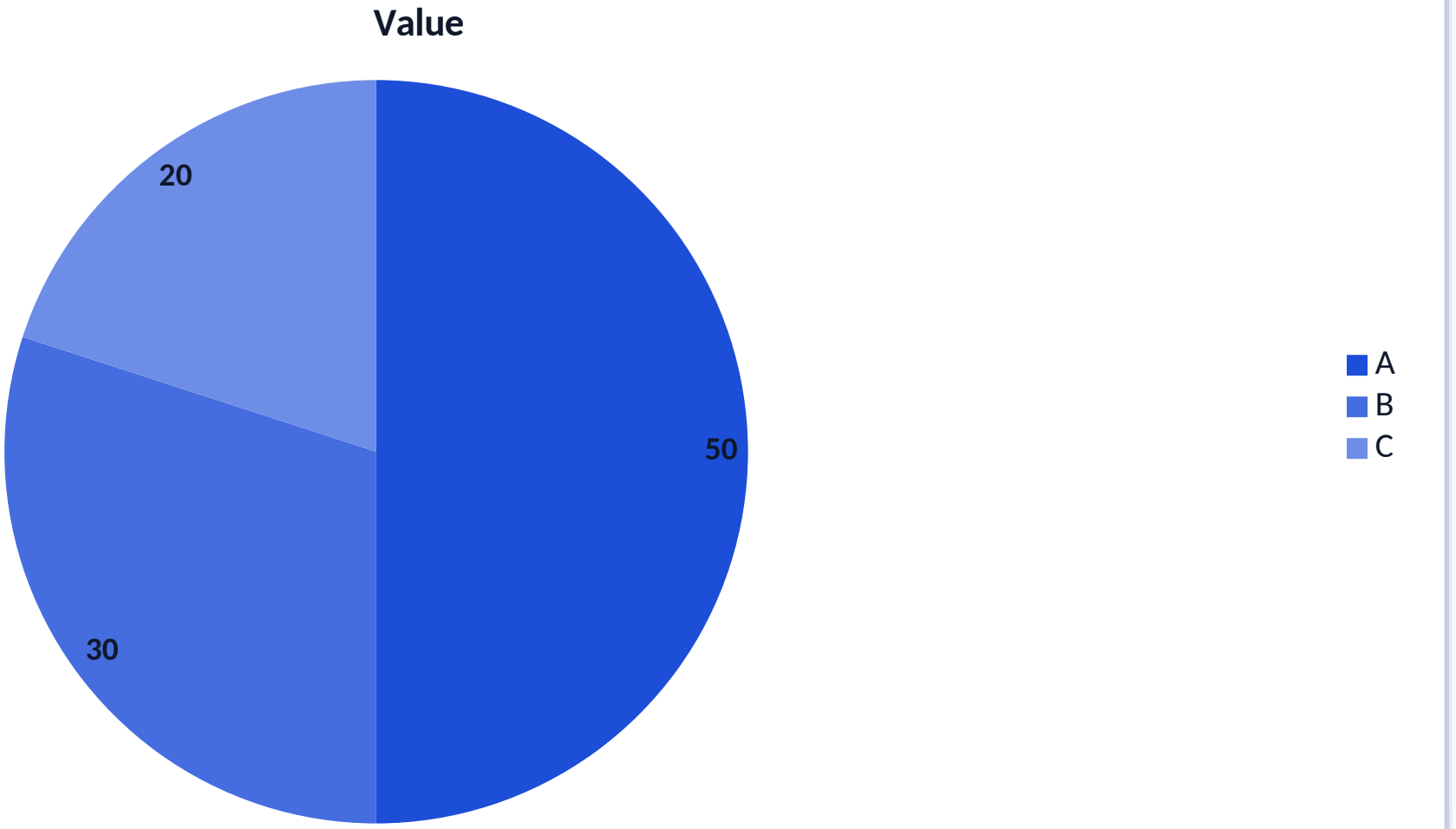
CHART_CATEGORICAL



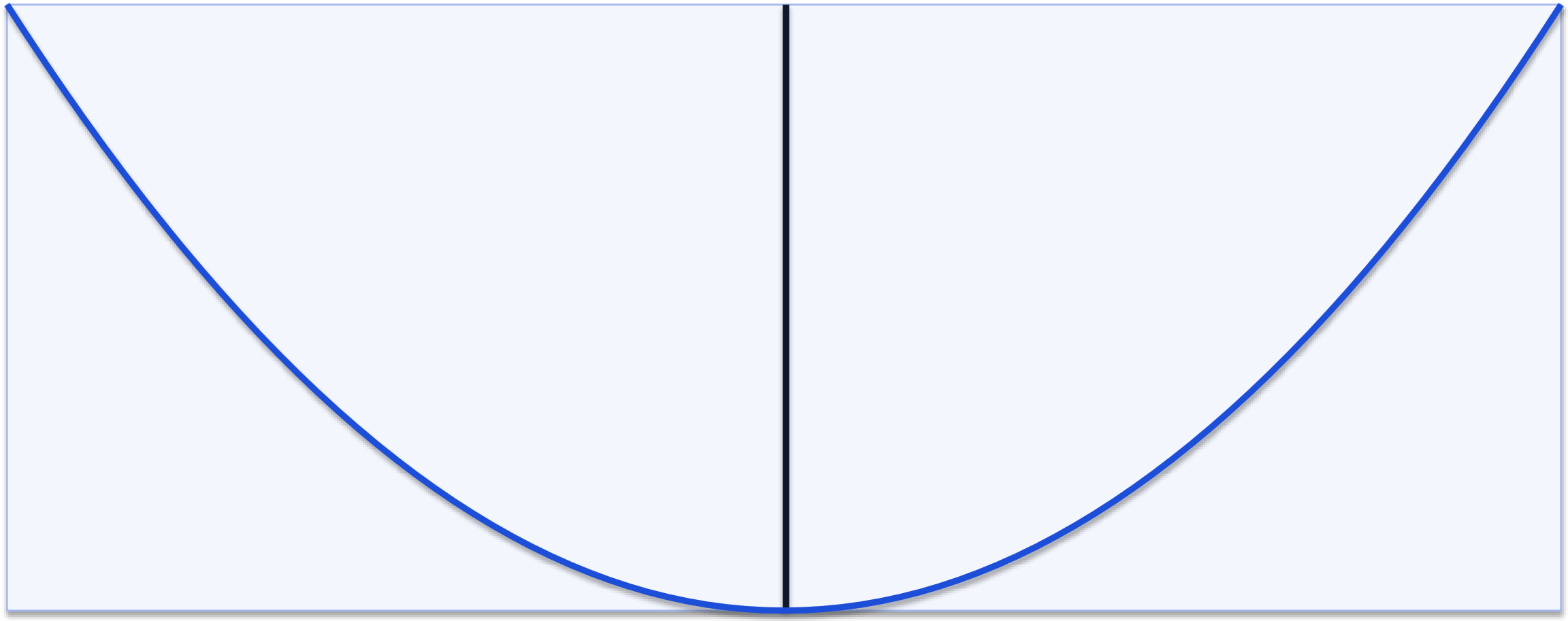
CHART_TREND



CHART_PART_TO_WHOLE

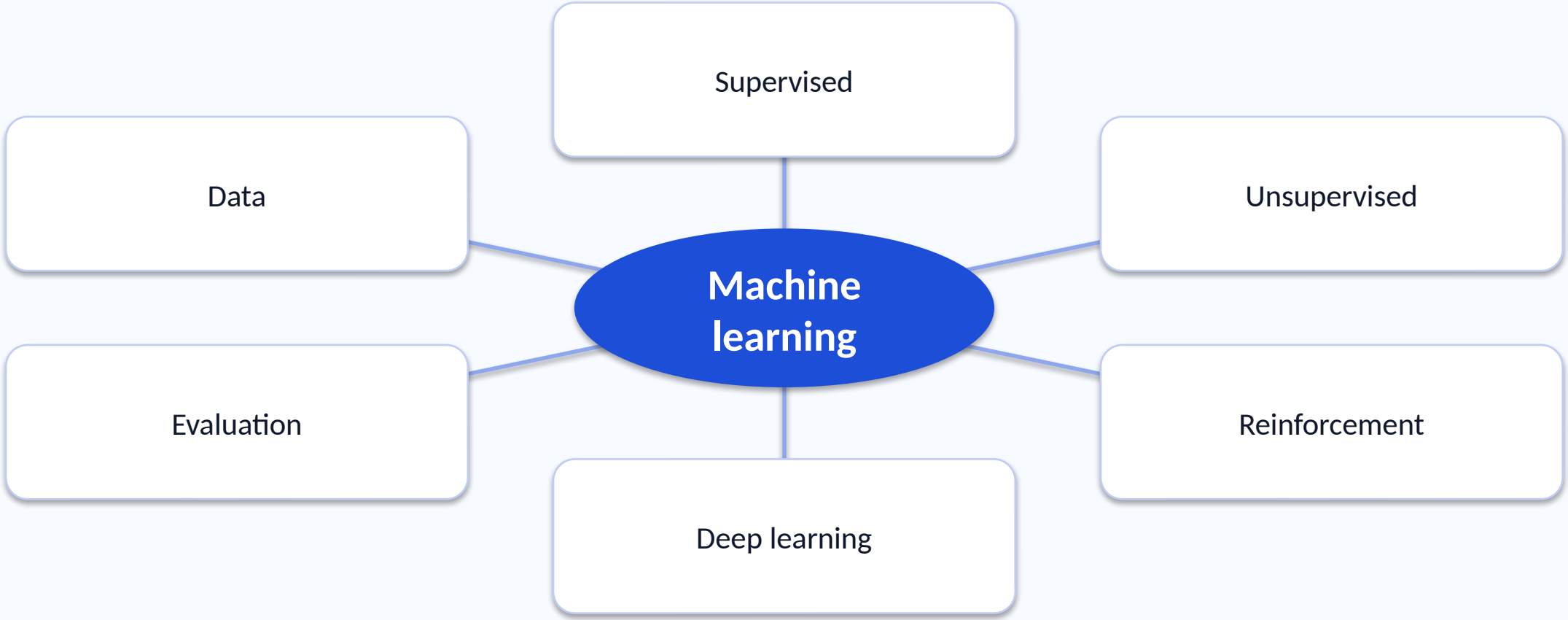


FUNCTION_PLOT



y = x**2

MIND_MAP



Key takeaway: **Key point here.**

KEY_POINTS

Scalability

Handles growth

Security

Protects data at rest and in transit with modern ciphers

Cost

Pay as you go

Speed

Low latency

Reliability

99.99% uptime

A short note about the slide topic.

CHECKLIST

1. Define the problem
2. Collect data
3. Split train/test
4. Train baseline
5. Evaluate
6. Iterate